



Nilesh Rohidas Barandwal
CSE(Computer Science & Engineering)
GitHub Profile

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Examination	University	Institute	Year	CGPA
Masters	IIT Dharwad	IIT Dharwad	2026	8.15
Graduation	Savitribai Phule Pune University	Pune Institute of Computer Technology	2021	7.99
Intermediate	MSBSHSE	Arihant College	2017	64.31%
Matriculation	MSBSHSE	St. Lawrence High School	2015	83.60%

WORK EXPERIENCE

- Avegen India Private Limited (Web Application Developer)** (Aug'21–Apr'22)
 - Collaborated with cross-functional teams to design, develop, and deliver scalable digital health products in an agile environment.
 - Built and maintained REST APIs and optimized data pipelines to improve performance and reliability.
 - Conducted sprint planning and tracking using JIRA, improving team efficiency and delivery predictability.
 - Designed test plans, debugged production issues, and authored technical documentation.
- Intellect Mind & Talent Management Innovations LLC USA (Full Stack Developer Intern)** (Aug'20–Jan'21)
(Guide: Mr. Ashwin Chaudhari and Dr.Gaurav Bansod)
 - Built responsive UI modules using React.js and developed an admin dashboard for web application management.
 - Developed secure, scalable backend APIs using Node.js and Express.js; managed database integration and authentication workflows.
 - Implemented user authentication and authorization workflows following security best practices.
 - Deployed and tested full-stack applications on cloud servers, troubleshooting issues to ensure reliability.
- IIT Bombay, E-Yantra (Summer Intern)** (May'20–Jun'20)
(Mentor: Prof. Kavi Arya, Dr.Rathin Biswas, Mr.Kedar Anavardekar)
 - Collaborated with cross-functional team to design and develop the MHRD-sponsored Rakshak App, working in an agile setup.
 - Developed reusable React.js UI components and integrated Google Maps API for real-time COVID data visualization.
 - Built clean and efficient REST APIs to enable reliable data exchange between system components.
 - Optimized real-time navigation algorithms to avoid unsafe or COVID-affected routes.
 - Maintained technical documentation and co-authored research paper based on project results.

SHIPPED PROJECTS

- Bolna Voice AI Lead Qualifier** Next.js 14, Express, SQLite, Bolna Voice AI
 - Agentic voice system that autonomously conducts outbound sales calls and qualifies leads as Hot/Warm/Cold without human involvement.
 - Implements multi-turn AI conversation, real-time webhook processing, and a live dashboard with 15s polling cycle.
 - Live: `bolna-lead-qualifier.vercel.app` Repo: `https://github.com/NileshBarandwal/bolna-lead-qualifier`
- AMM-DEX** Solidity, Hardhat, React, Ethers.js v6
 - Constant-product AMM ($x \times y = k$) with LP tokens, 0.3% fee accrual, and slippage protection deployed on Sepolia testnet.
 - Live: `amm-dex-orcin.vercel.app` Repo: `github.com/NileshBarandwal/AMM-DEX`
- VectorShift AI Pipeline Builder** React, Node.js, FastAPI
 - Visual LLM workflow builder enabling chaining of AI components including prompts, models, and data sources.
 - Repo: `github.com/NileshBarandwal/vectorshift-pipeline-builder`
- Armatrix Team Directory** FastAPI, Next.js, TypeScript
 - Team directory REST API with 17 pytest unit tests, deployed on Railway + Vercel.
 - Repo: `github.com/NileshBarandwal/armatrix-team`

ACADEMIC PROJECTS

- **Secure AI Execution Pipeline using ZKPs, TEEs, and Blockchain (M.Tech. MTP, IIT Dharwad)**
(Guide: Dr. Siba Narayan Swain) (MTP-1 Completed, MTP-2 In Progress)
 - Designed, developed, and deployed a secure AI execution architecture using Trusted Execution Environments (TEEs) to ensure isolated and tamper-resistant model inference.
 - Performed comprehensive threat modeling of execution-time risks, including adversarial manipulation, model extraction, backdoor triggers, denial-of-service, and side-channel attacks.
 - Built a scalable secure workflow with remote attestation, encrypted model and data handling, and controlled inference access across cloud environments.
 - MTP-2 (In Progress): Implementing a decentralized verification and attestation layer using Zero-Knowledge Proofs (ZKPs) and blockchain (Solidity smart contracts) to validate AI execution correctness without revealing sensitive data.
 - Evaluating security, scalability, and performance trade-offs under realistic adversarial scenarios.
- **SAMS (Student Attendance Management System) (CS618 MTech IITDH)**
(Guide: Ms. Gayathri Ananthanarayanan)
 - Collaborated with team to design and develop a full-stack web application using Django, MySQL, and FastAPI with clean, scalable REST APIs.
 - Implemented real-time barcode-based attendance with manual fallback mechanism; built responsive frontend using HTML/CSS/JavaScript.
 - Designed secure role-based access control (RBAC) for students, teachers, TAs, and admins with custom dashboards for marking, analytics, and reporting.
 - Developed normalized ER schema and optimized SQL queries for reliable data modeling, analytics, and reporting.
 - Prototyped UI in Figma, validated API endpoints via auto-generated Swagger docs, and added unit tests for reliability.
 - Repo: github.com/NileshBarandwal/CS618_MiniProject_Django

PUBLICATIONS

- **Journal Title:** New Techno Humanities. **Article Title:** A comprehensive easy-to-use techno-human solution for the resource-constrained nations in the fight against communicable diseases like the coronavirus pandemic.

TECHNICAL SKILLS

- **Languages:** C++, JavaScript, TypeScript, Python, SQL, Solidity
- **Frameworks & Tools:** Node.js, Express.js, FastAPI, Django, React.js, Next.js, REST APIs, Git/GitHub, JIRA
- **AI & ML:** LLM APIs, LangChain, Prompt Engineering, NumPy, Pandas, Scikit-learn (basic)
- **Databases & Cloud:** MySQL, PostgreSQL, SQLite, Docker, Vercel, Railway, Render
- **Web3:** Hardhat, Web3.js, Ethers.js, IPFS

POSITIONS OF RESPONSIBILITY

- **Senate & PGAPEC Student Member, IIT Dharwad (2025–26)** – Represented M.Tech. students in academic policy discussions, curriculum review, and student–faculty coordination.
- **Student Placement Coordinator, IIT Dharwad** – Coordinated placement activities, acted as liaison between students and recruiters, organized full-time job drives, and managed team logistics and communication.
- **Inbound Head, International Relations Office, IIT Dharwad** – Led selection and management of buddies for the inbound international student program; coordinated with IRO Head and office; organized events for international students.
- **Head TA, Computer Programming (CS101)** – Managed TAs, assisted professor in lab design, and evaluated coursework for 300+ students.
- **Teaching Assistant (DAA CS205/CS211, DSA CS201/CS205)** – Assisted professors in tutorials, labs, and student evaluations.
- **Head Associate, Web Team – HarshTaal '25 (IIT Dharwad Cultural Fest)** – Led team for website design, deployment, and maintenance.

ACHIEVEMENTS & ACTIVITIES

- **e-YIC (IIT Bombay, MHRD–NMEICT)** – Regional finalist; top 49 teams among 300+ nationwide.
- **e-YRC Robotics Challenge (IIT Bombay)** – Top 25 teams in India to successfully submit final solution.